

Perception Aware Dynamic Diffusion Transformers (PA-D²IT) for Safety Critical Scene Generation

Meenakshi Rajpurohit · Project Proposal Essay

Introduction

Diffusion models have become the dominant approach to high-fidelity image synthesis, and the Diffusion Transformer (DiT) family has shown that a transformer denoiser operating on latent patches scales more gracefully than the convolutional U-Nets. The Dynamic Diffusion Transformer (D²iT), treating every region of an image with identical generative capacity is wasteful. Some regions are visually simple and can be reconstructed from a coarse representation; others are visually complex and need finer treatment. D²iT therefore computes the Shannon entropy of the image latent, thresholds it to produce a "grain map" marking each region as coarse or fine, and lets a multi-grained transformer allocate tokens accordingly.

Our project begins from the observation that entropy is the wrong signal when the images are driving scenes. Shannon entropy is fundamentally a texture detector, it measures how busy the pixels are. Safety, by contrast, is a semantic property, and in driving scenes the two frequently point in opposite directions. A pedestrian in dark clothing at night occupies a smooth, low-contrast region: low entropy, but the single most important object in the frame. A textured tree line or a patch of foliage is the exact reverse: high entropy, and completely irrelevant to whether the vehicle brakes. Under entropy-driven allocation, D²iT spends its finest generative capacity on the tree line and treats the pedestrian as background.

We propose PA-D²iT (Perception-Aware D²iT), which replaces the entropy signal with a semantic importance field derived from the scene layout and the bounding boxes and class labels that already ship with every driving dataset. Everything downstream of that one block is inherited unchanged from D²iT.

Layout-conditioned methods like GeoDiffusion show generated frames can improve detectors, and adaptive-compute work already uses saliency signals but layout there steers content, not capacity, and saliency saves compute rather than concentrating it. Our contribution is the integration: importance-driven capacity allocation, evaluated on whether safety-critical objects actually improved.

Motivation

The cases that matter most in autonomous driving are long-tail cases: a pedestrian at night, a cyclist in rain, an occluded child near a parked car. These are simultaneously rare in real datasets, hard for detectors, and critically exactly the low-contrast, low-entropy regime that an entropy-driven allocator will systematically under-serve. The current approach therefore makes the problem worse precisely where it is already worst. If it works, the payoff is better synthetic

training data exactly where perception systems fail, since object-region fidelity unlike whole image aesthetics translates fairly directly into detector performance, benefiting AV perception teams needing long-tail augmentation and safety-validation work needing controllable rare events. Underneath sits a question that generalizes past driving: should a generative model allocate compute by texture or by meaning, when a few hundred pixels decide whether a car brakes?

Work Plan

Method. Scene layout becomes a tiered importance field vulnerable road users weighted highest, vehicles intermediate, static background lowest thresholded at a calibrated fine-grain ratio to yield the grain map that drives dynamic patchify, with layout also entering via cross-attention and training using D²iT's multi-grained loss scaled by a safety weight on VRU regions. Critically, the grain map derives from the layout rather than the noisy latent, making it deterministic at every timestep and sidestepping by construction the noisy-routing failure that degrades diffusion-MoE routers at high noise.

Datasets. Cityscapes is our primary development set, chosen for clean instance annotations, manageable scale, and standard baselines.

Models. We will use a DiT-B-class transformer of roughly 136M parameters performing latent diffusion over a frozen $f=8$ VAE, with DDPM training and DDIM sampling.

Metrics. Our two primary metrics are object-region FID, computed on cropped VRU regions rather than whole frames, and downstream detector mAP, obtained by training a detector on generated data and testing it on real data.

Baselines. We will compare against DiT-B (no layout, uniform capacity), DiT-B + layout (layout-conditioned but still uniform capacity, isolating the value of conditioning alone), and D²iT-B (entropy-driven grain the real baseline to beat, since it shares our machinery and differs only in signal). Every model is trained at matched compute: this is a proof of concept about allocation, not a bid for state of the art.

Risk Analysis

The main risk is that importance simply loses to entropy; that texture is a good enough proxy after all so we have designed the matched-compute comparison to isolate the signal and will ablate both the safety weight and the fine-grain ratio, ensuring even a negative result is interpretable. Global FID may also regress as capacity shifts toward objects, which is why we keep it as an explicit guardrail and will report the trade-off curve rather than a single favorable point. Compute limits are handled by freezing the VAE and keeping every run inside 12–16 GB, and novelty overlap by positioning on allocation rather than conditioning, with the DiT-B + layout baseline there to demonstrate the difference.

Conclusion

The proposal is reduced to a single sentence: put the model's capacity where the risk is. If the hypothesis holds, we gain a principled way to direct generative compute by semantics rather than texture. If it fails, the matched-compute ablations will tell us why entropy was sufficient which is itself a result worth reporting.

References

[1] <https://www.coursera.org/projects/how-diffusion-models-work-project>

[2] Zhao, W., Han, Y., Tang, J., Wang, K., Song, Y., Huang, G., Wang, F. and You, Y., 2025, May. Dynamic diffusion transformer. In *International Conference on Learning Representations* (Vol. 2025, pp. 65520-65552).

[3] Chen, K., Xie, E., Chen, Z., Wang, Y., Hong, L., Li, Z. and Yeung, D.Y., 2024, May. Geodiffusion: Text-prompted geometric control for object detection data generation. In *International Conference on Learning Representations* (Vol. 2024, pp. 8979-9001).

Appendix

Prompt:

I have attached my essay here could you help me with grammatical mistakes and make this essay more easy to read and understandable able format. Plus use less token as you can so give me exact places needs change.

Strategized targeted feedback approach to minimize token usage

Strategized targeted feedback approach to minimize token usage

Here are the specific fixes needed:

Grammar/punctuation fixes:

1. "The Dynamic Diffusion Transformer (D²iT), makes an important observation" → remove comma after "(D²iT)"
2. "Some regions are visually simple... others are visually complex and need finer treatment" — fine as is.
3. "A pedestrian in dark clothing at night occupies a smooth, low-contrast region: low entropy, but the single most important object in the frame." → add comma: "region: low entropy, **but is** the single most important object..."
4. "layout there steers content, not capacity, and saliency saves compute rather than concentrating it" — add "but" for flow: "layout there steers content, not capacity, **and** saliency..." (currently fine, optional)
5. "Cityscapes is our primary development set, chosen for clean instance annotations, manageable scale, and standard baselines." — fine.

6. "vulnerable road users weighted highest, vehicles intermediate, static background lowest **thresholded** at a calibrated fine-grain ratio" → missing comma/connector: "static background lowest, **then** thresholded at..."
7. "D²iT-B (entropy-driven grain **the real baseline** to beat, since it shares our machinery and differs only in signal)" → insert em dash or comma: "entropy-driven grain**, ** the real baseline to beat"
8. "that texture is a good enough proxy after all **so** we have designed" → needs comma/period: "after all. **So** we have designed..." or "after all**, ** so we..."
9. "which is itself a result worth reporting" — fine.

Readability suggestions (structural, not grammar):

- Motivation paragraph is one long run-on sentence ("The current approach therefore makes the problem worse... benefiting AV perception teams...") — split into 2–3 sentences.
- Risk Analysis paragraph merges four separate risks into one sentence chain — break into a short list or separate sentences (one per risk: signal risk, FID regression, compute limits, novelty overlap).

Want me to produce the fully corrected version as a doc, or is this list enough to edit yourself?